Varad Rane

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Summary

Backend engineer, 3 years production experience across trading infrastructure and applied LLM systems. Built token-based auth, an API gateway and production monitoring for trading APIs serving 2M users at one of India's largest brokerages. At a healthtech startup, shipped an LLM document-extraction pipeline that automates ~95% of client intake and a local-first RAG pipeline adopted for 97% of new client entries, then took the RAG system to production on Azure as an agentic assistant. Also built a serverless AWS extraction pipeline on Step Functions, Bedrock and Aurora/pgvector with full Terraform IaC.

Experience

Software Developer Feb 2026 - Present
Field Worker (Contract Position) Princeton, NJ

- Built an LLM document-extraction pipeline (GPT-4.1) turning multi-column, cross-page PDF reports into schema-validated JSON, **automating ~95% of client intake** and removing manual data entry from the workflow.
- Architected a local-first RAG pipeline for clinical case notes (Ollama, ChromaDB, sentence-transformer embeddings, PostgreSQL vector storage) engineered to **run inside an 8GB VRAM budget** so sensitive data never left the environment; adopted for **97% of new client entries in production**.
- Took a local RAG prototype to production on Azure: an agentic case-summary assistant inside the Node.js API (Azure AI Foundry GPT-4.1, PostgreSQL/pgvector) with 10 tool-calling functions for per-patient summaries, risks, open issues, urgent flags and caseload-wide reviews, plus its React chat UI.
- Built a nightly QA compliance sweep pairing deterministic SQL checks for unit and form gaps with an LLM verdict on whether each task narrative justifies what was billed; the model never does arithmetic, and results persist for a compliance dashboard.
- Designed a reproducible benchmarking harness measuring latency, throughput and retrieval quality across chunking, retrieval and quantization strategies; heading-based markdown chunking won, **raising retrieval accuracy 72% to 93%**.

Teaching Assistant Sep 2025 - Dec 2025
New Jersey Institute of Technology Newark, NJ

- Built an automated grading system running student Python and C++ submissions against test suites in isolated containers, with an LLM generating structured HTML feedback on code quality; cut **turnaround for 50 students from 5 days to 6 hours**.

Software Engineer II Jun 2023 - Aug 2024
Kotak Securities Mumbai, India

- Built **token-based authentication from scratch** for trading APIs serving 2M users at **~260 req/s peak, 30% above** the previous 200 req/s, setting the security and audit-compliance baseline for all later internal applications, **remediated 12 penetration-test findings** and authored the compliance docs.
- Architected alert management across **100+ production servers and message queues** with configurable health checks, cutting quarterly downtime from **6 hours to 5** and automating post-incident root-cause reporting.
- Built a .NET Standard 2.1 shared library for data access and logging adopted by **50+ engineers across 4 teams**, removed two network hops per request and resolved a production IIS socket-exhaustion failure.
- Designed an internal API gateway **centralizing auth, CORS and rate limiting, with SQL-driven routing** that enabled parallel live-environment testing against production data with no separate staging deployment.

Software Engineer Jun 2022 - Jun 2023
Kotak Securities Mumbai, India

- Optimized high-frequency production SQL via execution-plan analysis and indexing, **cutting p95 latency** on critical retrieval paths from **237 ms to 145 ms.**
- Migrated a legacy CRM (70 views) from **.NET Framework to .NET 7** with zero production downtime, resolving breaking changes and realigning auth flows to updated security standards.
- Built a **reusable bulk file-upload module** with validation, transactional error handling and user-facing recovery, **standardizing CSV ingestion** across 7 teams and halving development time for new upload workflows.

Projects

Cloud Document Extraction Pipeline
https://github.com/varadrane7/cloud-extraction-pipeline
Python, AWS Lambda, Step Functions, Bedrock, Aurora + pgvector, Terraform

- S3, Step Functions (Distributed Map), 8 Lambdas, Bedrock (Claude 3.5 Sonnet vision + Titan Embeddings), Aurora + pgvector

PrivGuard Sentinel

https://blog.varadrane.com/posts/privguard-sentinel

Python, GitHub Actions, LLM APIs

- LLM pull-request reviewer posting inline comments from structured JSON. Chunks diffs past model context limits.

NJ Transit Bus ETA Service

Python, GTFS-Realtime (protobuf), Docker, Home Assistant

- Real-time NJ Transit ETA service (Python, GTFS-RT, Docker): realtime → GPS-projection → timetable fallback with explicit staleness states; feeds Home Assistant alerts.

Education

New Jersey Institute of Technology Newark, NJ

MS, Computer Science · Jan 2025 - May 2026 · GPA 3.90 / 4.00

University of Mumbai Mumbai, India

BE, Electronics Engineering · Jun 2018 - May 2022 · GPA 9.32/10.0

Skills

Languages: Python, Go, C#, SQL/T-SQL, TypeScript, JavaScript, C++

Backend: REST API design, API gateways, microservices, distributed system design, OAuth 2.0, JWT, 2FA, IAM, rate limiting, ASP.NET Core, .NET Core/Standard, Node.js, GTFS-Realtime, protobuf, node-schedule

Data: PostgreSQL, SQL Server, pgvector, ChromaDB, query optimization, execution-plan analysis, indexing, schema migration

Cloud: AWS (Lambda, Step Functions, Bedrock, Aurora, S3, API Gateway, CloudWatch), Terraform, Docker, GitHub Actions, CI/CD, Linux, observability, Azure (AI Foundry, Container Registry, Web Apps), Microsoft Entra ID / MSAL

AI / LLM: RAG, embeddings, semantic search, vector DBs, prompt orchestration, structured output, LLM evaluation & benchmarking, chunking, quantization, Ollama

Other: React, Next.js, Angular, system design, code review, unit & integration testing, Agile

Certificates and Publications

AWS Building Production-Ready AI Agents with Amazon Bedrock AgentCore (2026)

Rane, V., et al. "Automated Crack Severity Level Detection and Classification for Surface Crack Using Deep Convolutional Neural Networks." Springer: Advanced Computing and Intelligent Technologies, Aug 2022